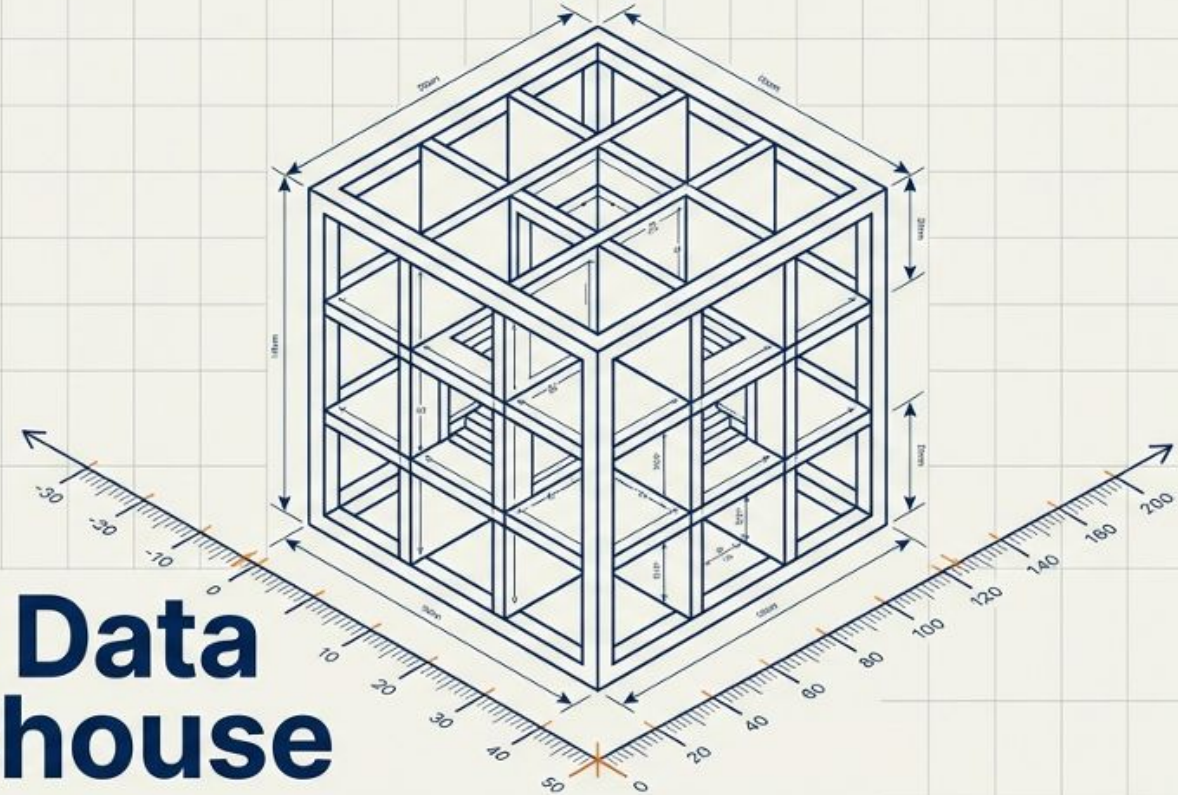


Olist Data Lakehouse

Architectural Blueprint & Business Intelligence Engine

Olist E-Commerce Data Lakehouse

Project Oveview

📦 Strategic E-Commerce Analytics via Google Cloud BigQuery & Modern Data Stack

Transforms fragmented Brazilian e-commerce transaction logs into a high-performance **Star Schema** for executive reporting. By automating the end-to-end flow from raw data to BI, we eliminated 100% of manual data preparation, providing a "Single Source of Truth" for Olist's 9+ disparate datasets.

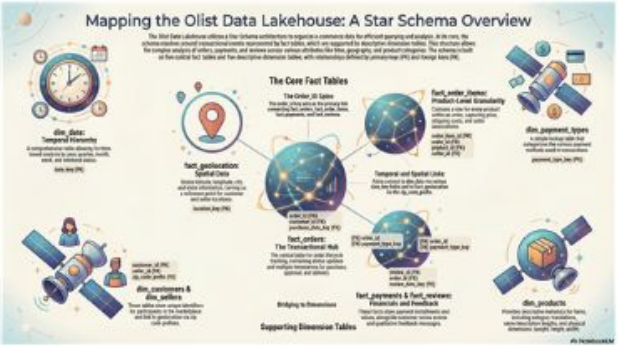
🛠️ Methodology: The Medallion Architecture

Implemented a modular **ELT (Extract, Load, Transform)** pipeline centered on data reliability and scalability:

Ingestion (Bronze): Automated raw data extraction via **Meltano** into **Google BigQuery**, preserving full data lineage.

Transformation (Silver & Gold): Fact/Dimension models, leveraged **dbt** to normalize schemas, standardize timestamps, and enforce **79+ automated data quality tests**.

Analytics (Rose): Developed denormalized **Data Marts** optimized for low-latency querying and executive-level visualization.

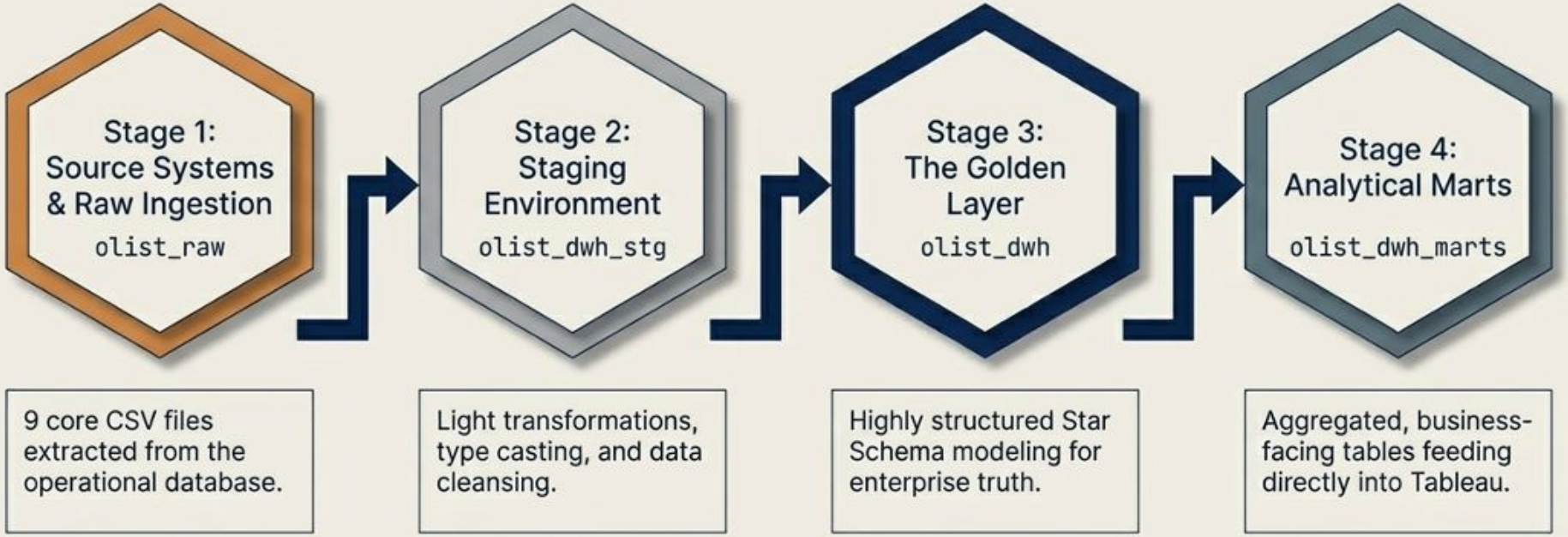


Quality & Trust: Integrated **Great Expectations** for data contract validation and **Tableau** for high-fidelity KPI tracking (e.g., Year-over-Year growth).

🚀 Key Insight: August 2018 Performance

The final dashboard reveals a **-4.13% YoY revenue dip** in August 2018 despite stable order volumes (**7,152 units**). This engineered pipeline allows leadership to instantly pivot from *observing* trends to *investigating* root causes in the underlying product categories.

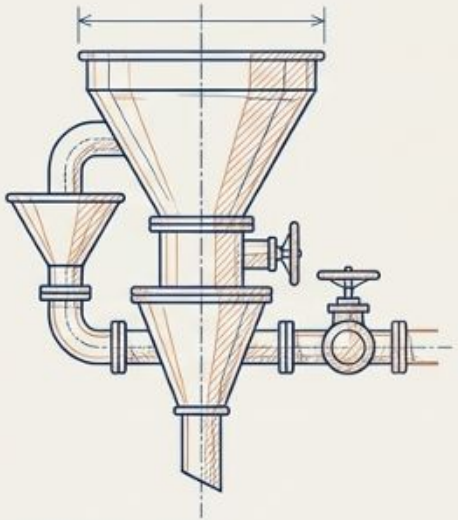
The Modern Data Stack Architecture



Design and Planning

Pipeline Mechanics: Ingestion to Transformation

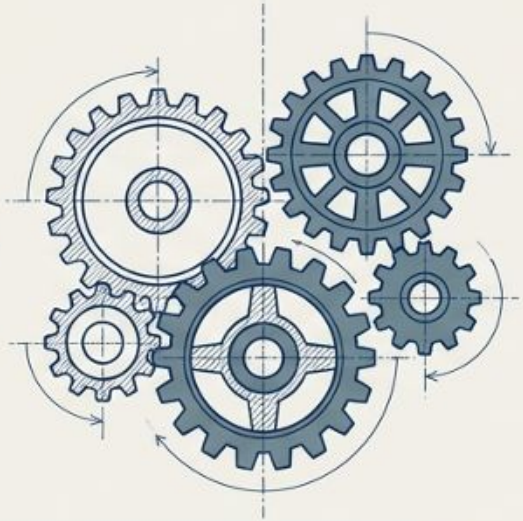
Meltano



Handles **extraction, loading**, and **orchestration**. Ingests 9 heavy operational files directly into the olist_raw layer.

raw_sellers	(174.7MB)
raw_geolocation	(61.27MB)

dbt (Data Build Tool)

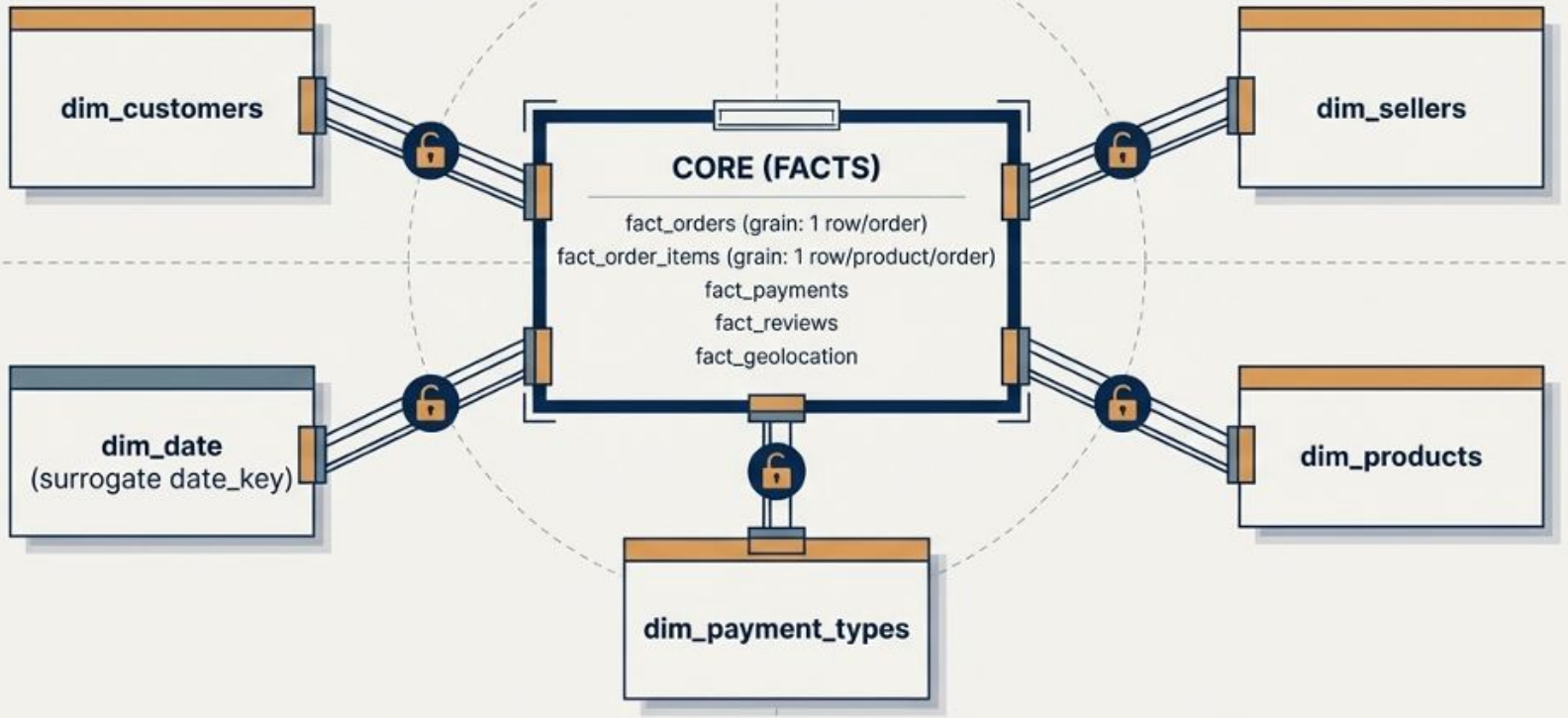


Executes rigorous, **version-controlled SQL transformations.** Standardizes strings, manages surrogate keys, handles safe casting, and tests data integrity from staging up to the analytical marts.

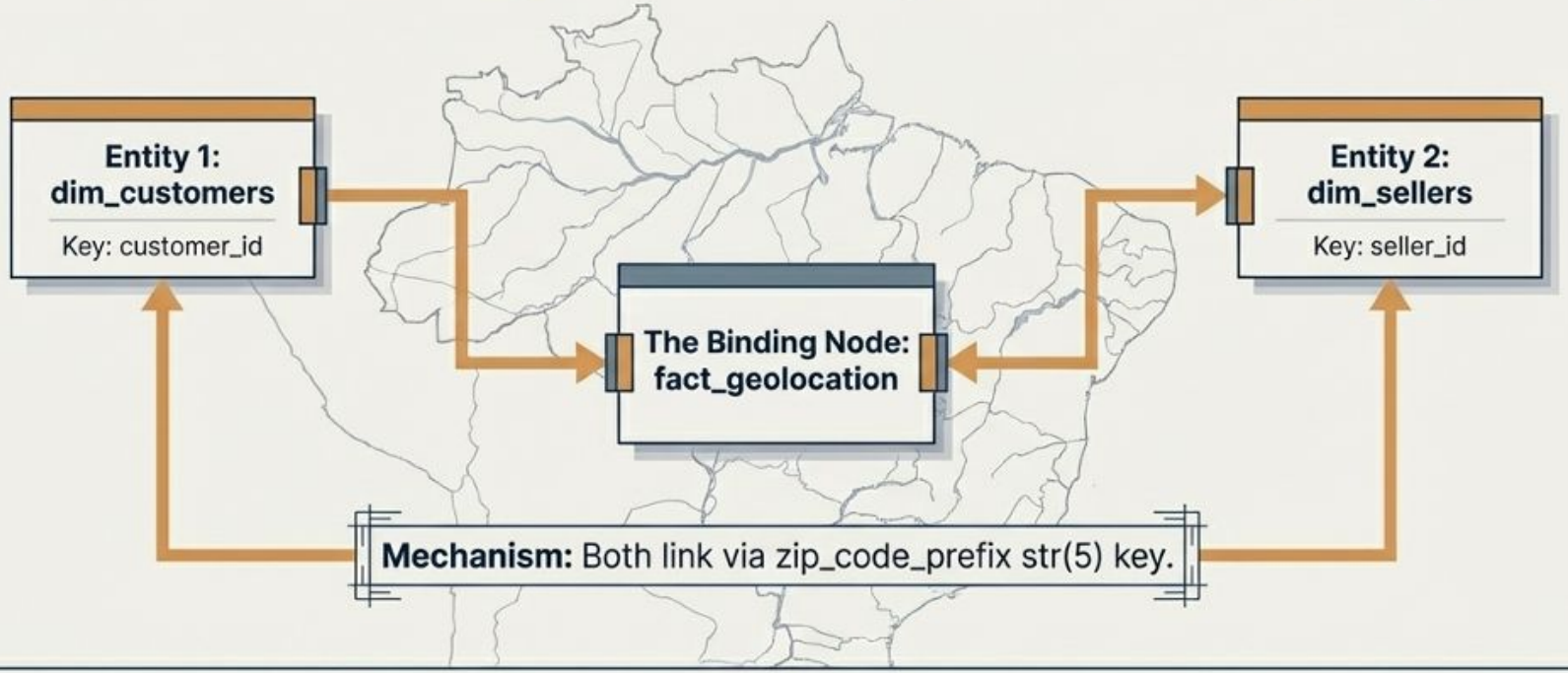
LOWER()
UPPER()

The ‘Golden Layer’ Star Schema

A fully normalized, single source of truth for the Olist enterprise.

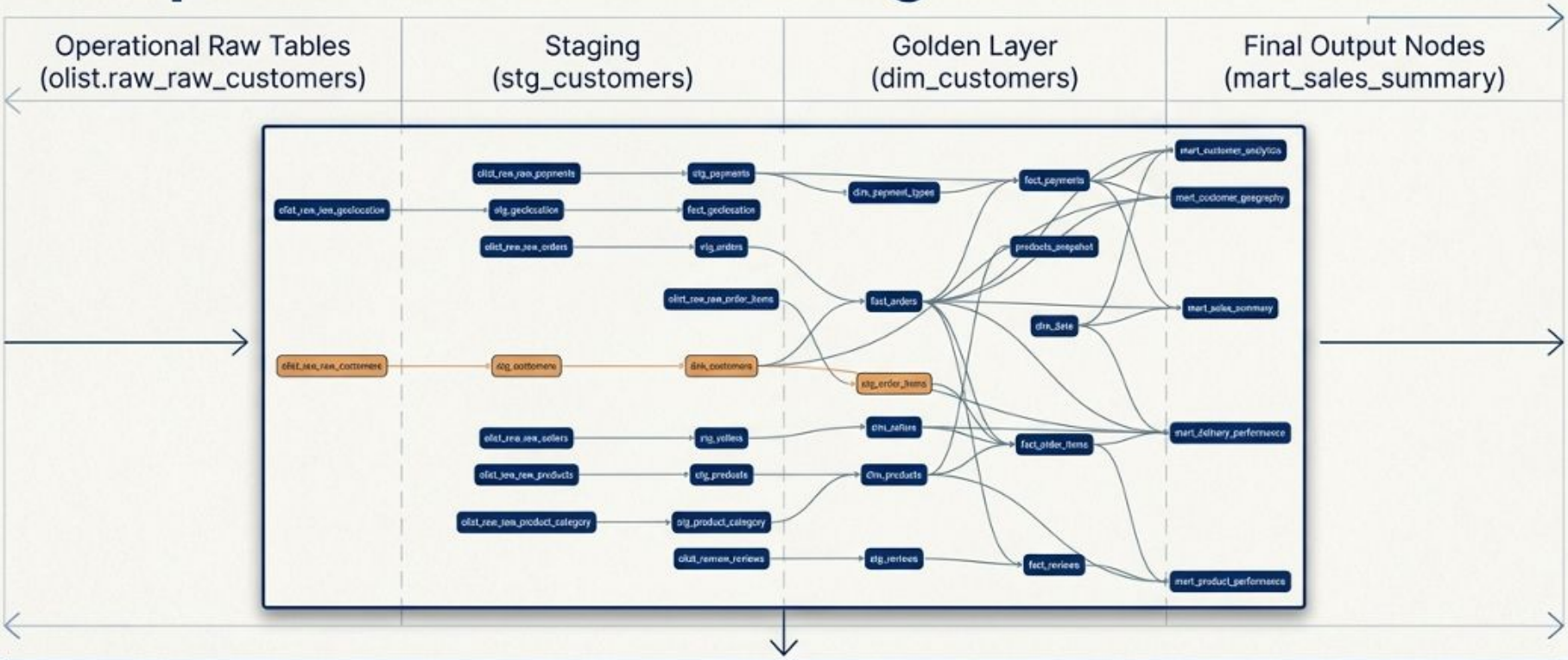


Geospatial Intelligence & Logistics Mapping



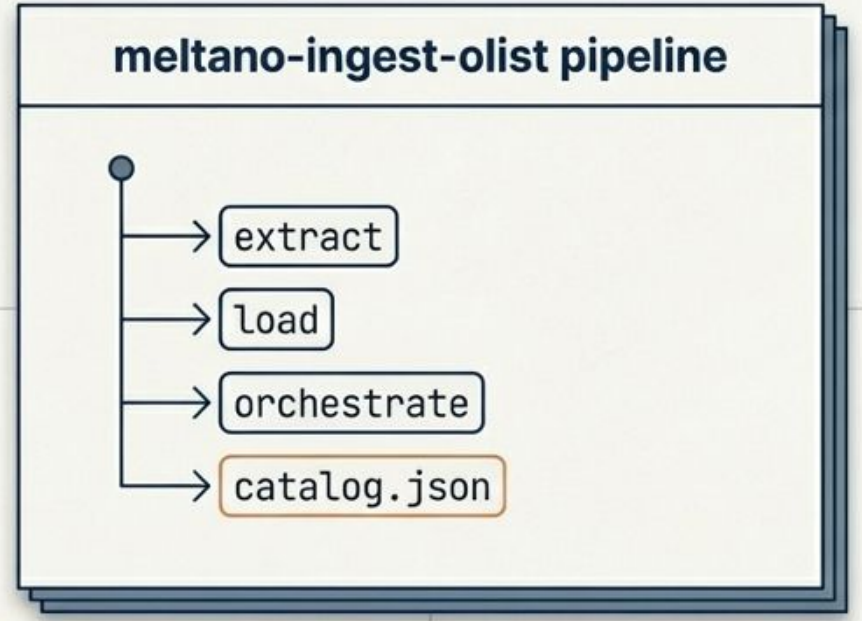
Business Impact: This exact relational join unlocks the ability to map delivery heatmaps, calculate shipping durations, and analyze cross-state commerce between buyers and sellers.

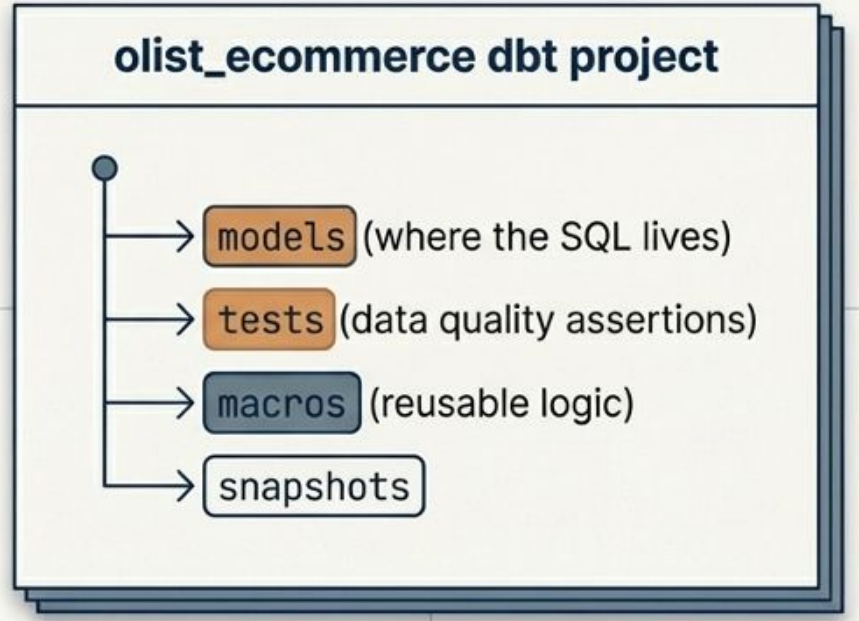
Transparent Data Flow & Lineage



Takeaway: Every metric on the BI dashboard can be traced backward through this lineage graph to its exact operational source, guaranteeing total auditability and isolating pipeline errors instantly.

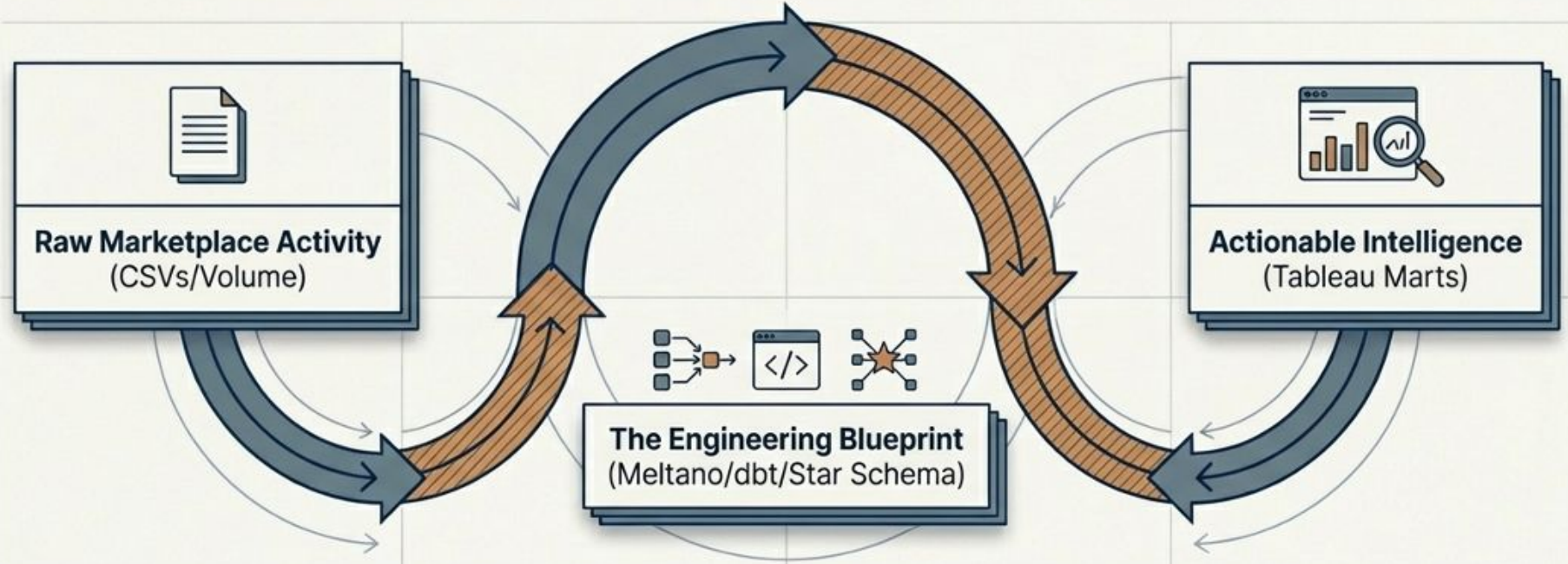
Project Structure & Engineering Rigor





Takeaway: The lakehouse is treated as a software product—fully version-controlled, strictly tested, and consistently logged.

Synthesis: The Lakehouse Value Engine



Core Insight: The Olist Lakehouse is not just a storage solution; it is an active translation engine. By applying rigorous software engineering principles to messy operational data, we bridge the gap between backend chaos and front-end clarity, enabling scalable, governed analytics.

Rose Mart Preview

Olist E-Commerce Data Lakehouse

Why is Rose Data Marts?

'**Centrol of True**' Golden Data Lakehouse to Actionable Business Ready - **Rose Marts**

1. The Challenge (The "Before")

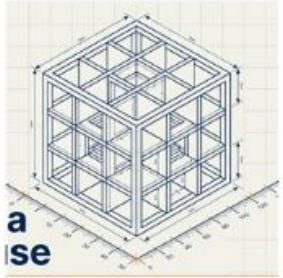
Data Silos: Raw data is scattered and inconsistent.
High Latency: Complex queries on raw data are slow and expensive.
Metric Drift: Different teams reporting different "Revenue" numbers.

2. Strategic Benefits of Data Marts

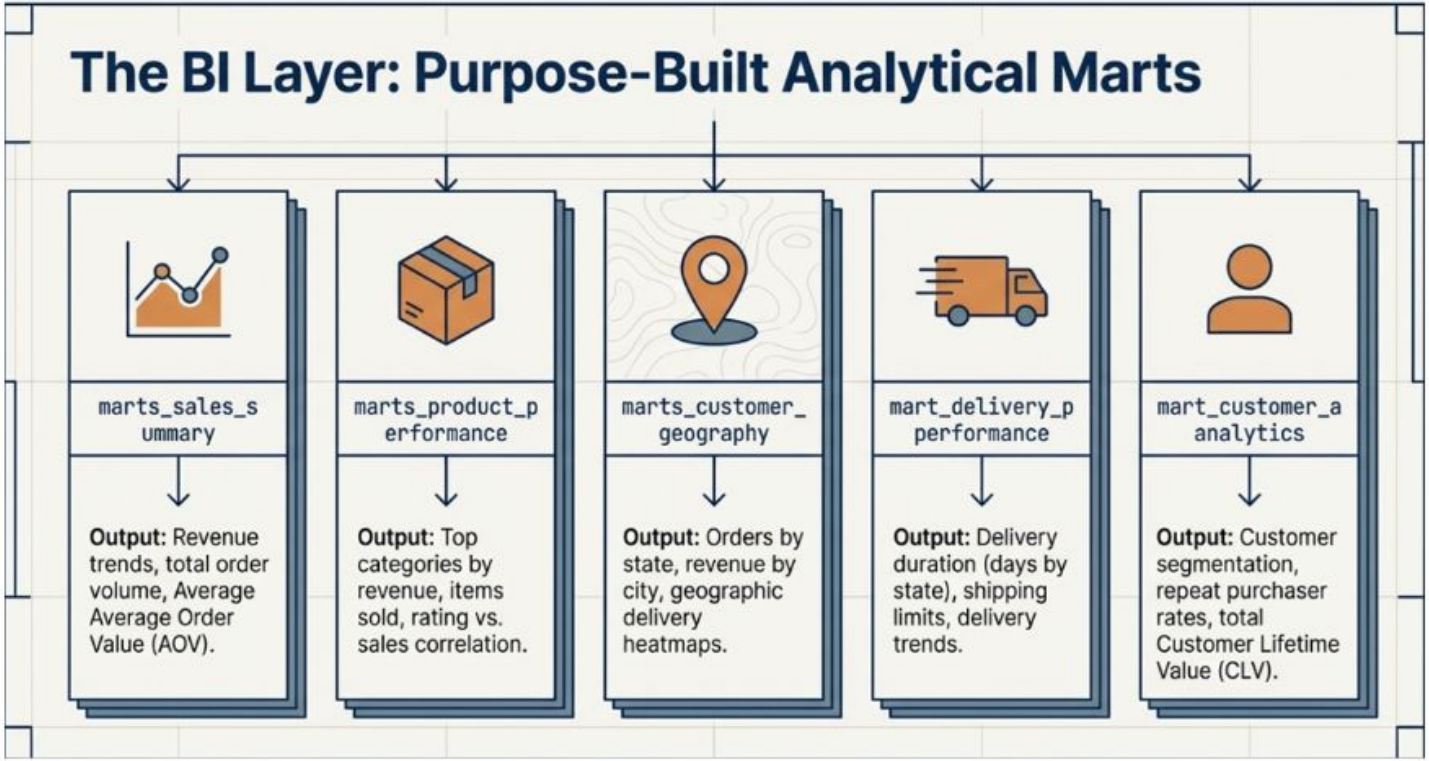
Performance: Pre-aggregated tables reduce dashboard load times by up to 80%.
Governance: Fine-grained access control (Role-Based Access) ensures data security.
Consistency: Shared definitions for KPIs across the entire organization.

3. Key Outcomes - Rose Marts

Cost Efficiency: Lower compute costs by querying smaller, modeled datasets.
Self-Service BI: Analysts can build reports without needing complex SQL joins.
Reliability: Automated testing in the transformation layer (e.g., dbt tests) ensures data quality.



What is Rose Marts?



How to use it?

Executive Sales Overview (Jan 2017~ Aug 2018)

Note: Results based on sample data of 5000.

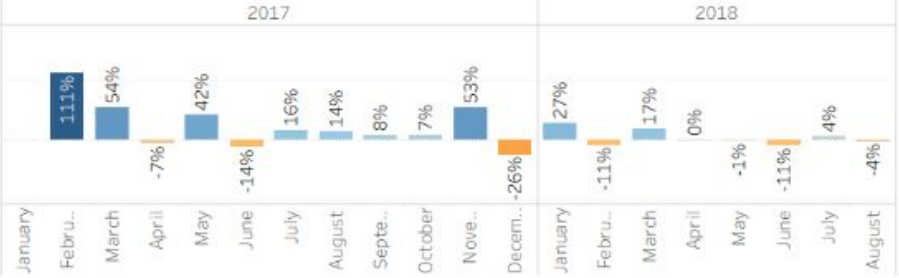
Total Revenue	Revenue MoM %	Total Order Volume	Avg. Order Value
$8.7M	▼ 4.1%	99K	$156
v.s yrs17 (20.1%)	(Aug v.s July 2018)		

Revenue Seasonality (Sales Heatmap)

Dark blue Indicated months were the "hottest".

Revenue MoM Growth %:

Spikes in Nov 2017 (Black Friday).



Average Order Value

Spikes in late 2017 (Black Friday), and Sept 2018

Revenue by Day of Week (2018)

People spend more on weekdays vs. weekends.

The Foundation is Set

1

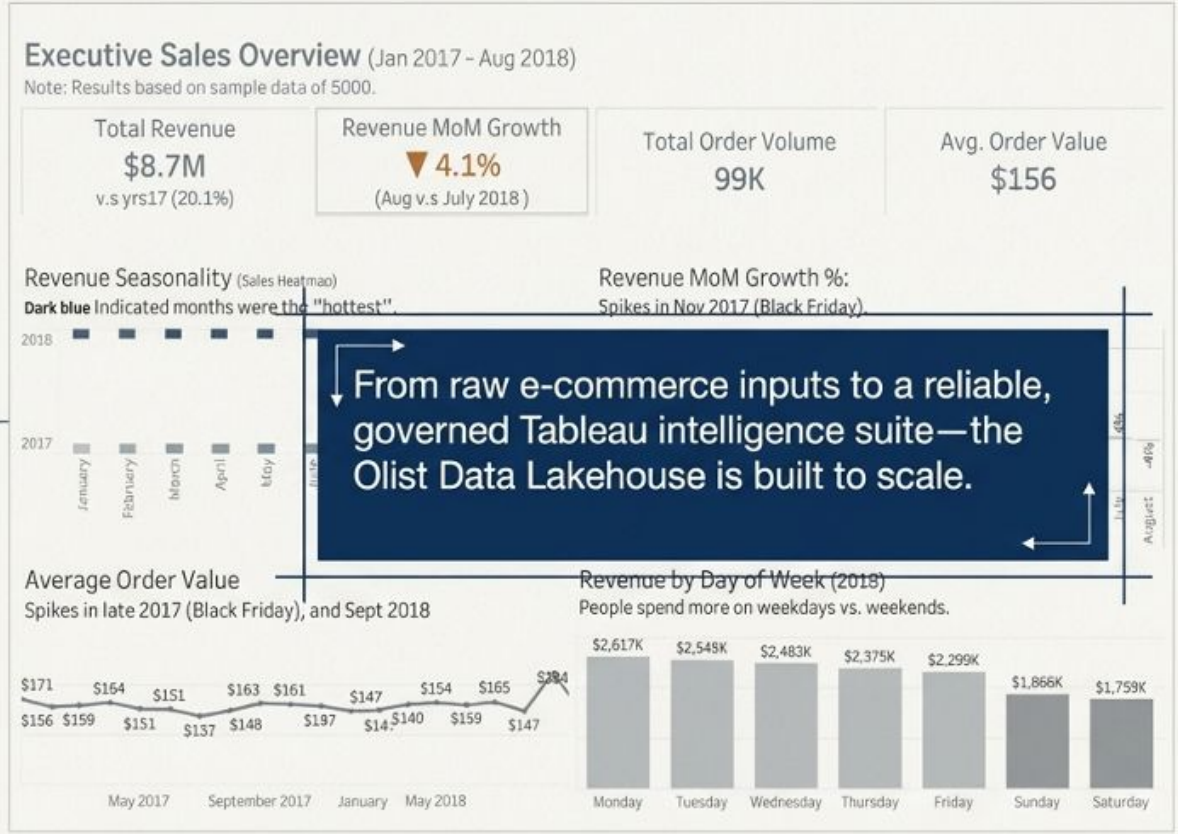
Automated, reliable ingestion pipeline.

2

Governed, test-driven Star Schema.

3

Five dedicated business-intelligence marts.



Q&A

Olist E-Commerce Data Lakehouse

That concludes my walkthrough of the Olist E-Commerce Data Lakehouse. I've documented the full ELT logic and dbt test results in my GitHub repository, which you can see linked here.

Thank you for your time—I'm now happy to take any questions.